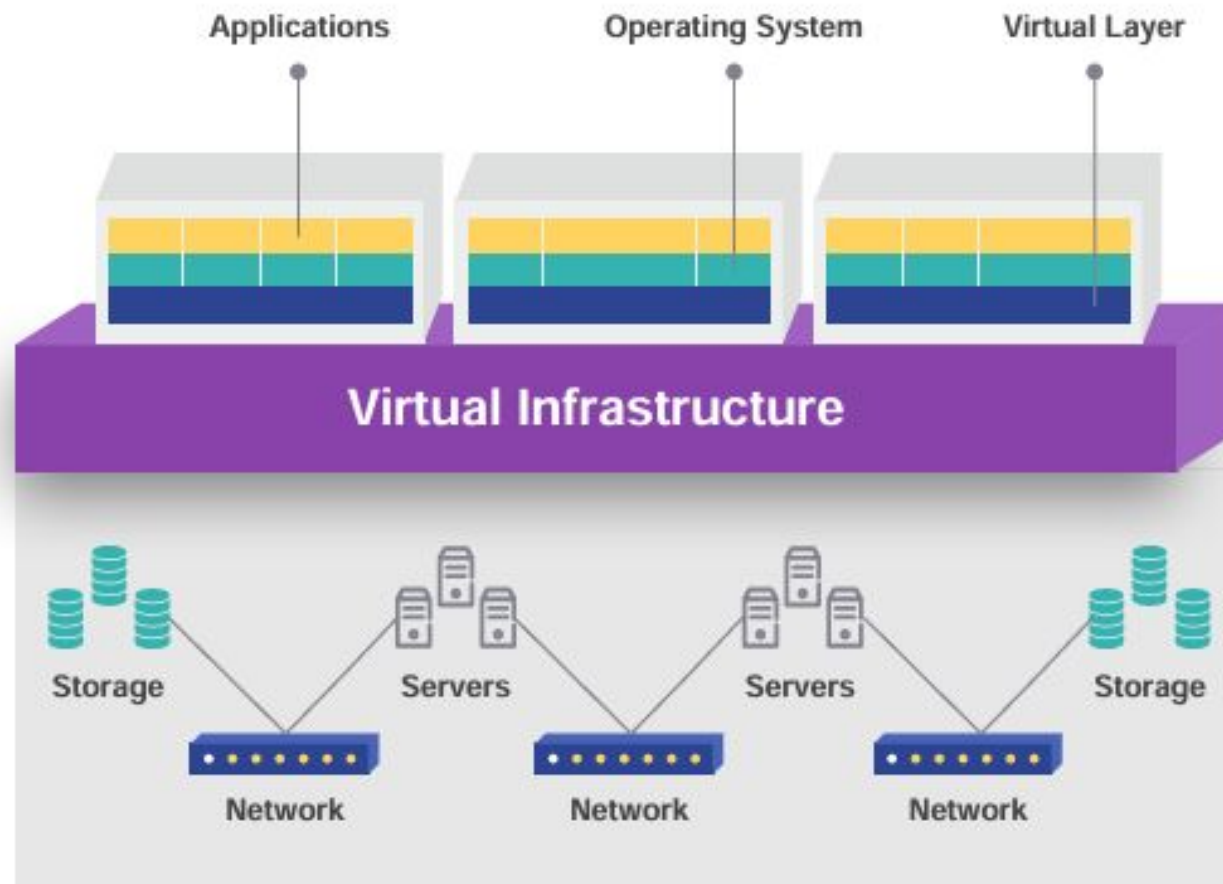


Recap – Virtualization



Visualizing Virtualization



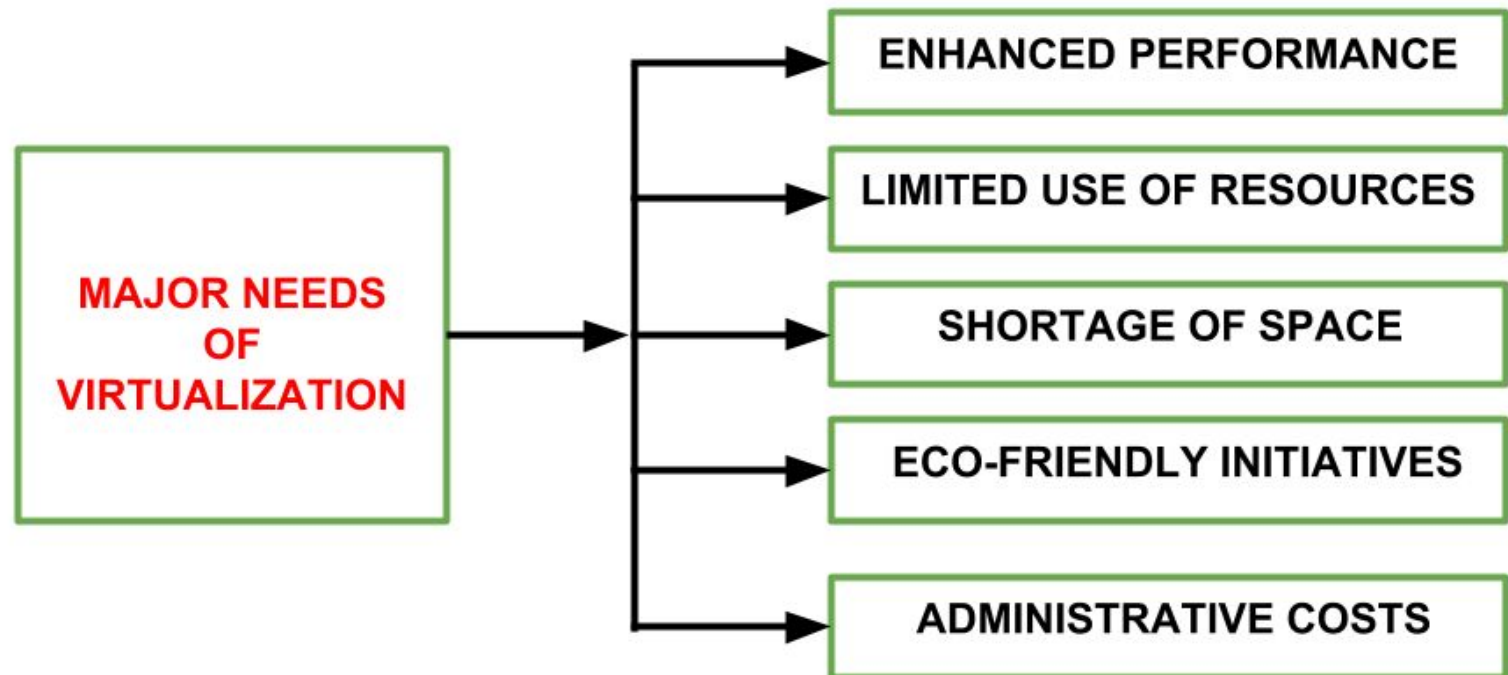
Virtualization Characteristics

Characteristics of Virtualization

- ***Partitioning:*** In virtualization, many applications and operating systems (OSes) are supported in a single physical system by *partitioning* (separating) the available resources.
- ***Isolation:*** Each virtual machine is isolated from its host physical system and other virtualized machines. Because of this isolation, if one virtual instance crashes, it doesn't affect the other virtual machines. In addition, data isn't shared between one virtual container and another.
- ***Encapsulation:*** A virtual machine can be represented (and even stored) as a single file, so user can identify it easily based on the service it provides. In essence, the encapsulated process could be a business service. This encapsulated virtual machine can be presented to an application as a complete entity. Therefore, encapsulation can protect each application so that it doesn't interfere with another application.
- ***Resource Abstraction:*** Virtualization enables the abstraction of physical resources such as processors, memory, storage, and network components. This abstraction allows multiple virtual instances to run on a single physical machine, optimizing resource utilization and maximizing efficiency.

Requirements of Virtualization

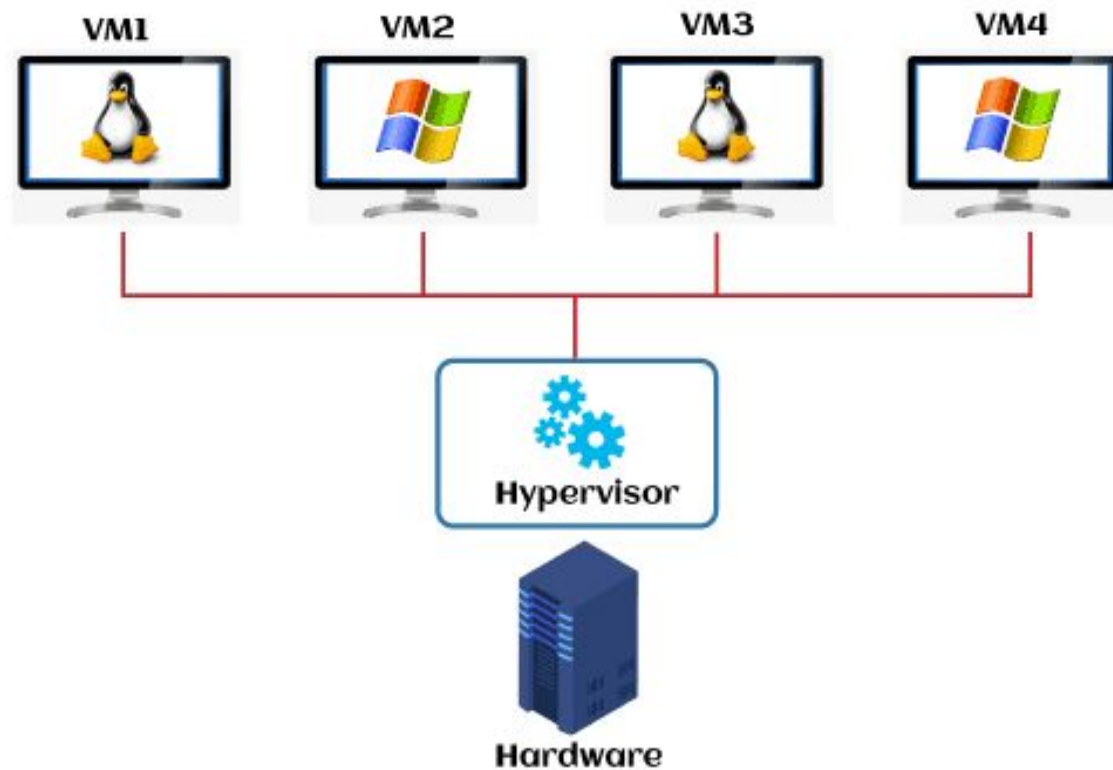
- **Objectives of Virtualization:** It enables creation of an intelligent layer of abstraction to hide the intricacies of the software or hardware that is underlying beneath the layer.



Advantages of Virtualization

- Reduced capital and operating cost
- Minimized or eliminating the downtime
- Faster Backup
- Seamless Migration of Resources
- Increased Efficiency and Productivity
- Failure Protection

Hypervisor in Virtualization



Hardware Abstraction

- Hardware abstraction is a key part of virtualization, which is the process of separating a computer's physical components into virtual computers, or virtual machines (VMs).
- The hardware abstraction layer (HAL) is a software component that acts as an interface between the hardware and the operating system, allowing VMs to interact with virtualized hardware devices.

Hardware Abstraction

Standardized interface

- The HAL allows VMs to interact with virtualized hardware devices using a standardized interface, regardless of the physical hardware on the host machine.

Migration

- VMs can be migrated between different physical hosts without requiring modifications to the guest operating system or applications.

Improved hardware utilization

- Virtualization can improve hardware utilization and system efficiency.

Flexibility

- Virtualization can provide flexibility in application deployment and protection.

Foundational Issues

- Technical or Business Issues
- Determine the hardware needed per demand
- Adding new layer in between the actual hardware and the virtual computers may lead to a little overhead which affects virtualized environments' overall performance.
- Virtualized environments rely on the software known as hypervisor. Any problems or vulnerabilities with the hypervisor could affect the virtualized infrastructure.
- Virtualization increases system stability. But, it creates a single point of failure in the hypervisor. All virtual machines executing on that host are impacted if the hypervisor malfunctions.

Software Provisioning

- There are several software options for provisioning in virtualization, including



VMware vSphere

A server virtualization platform that helps organizations manage virtual machines at scale.



Hyper-V

A type-1 hypervisor that allows users to create and manage multiple virtual machines (VMs).

Software Provisioning



Citrix Hypervisor

A virtualization management platform that helps organizations consolidate and contain their workloads.



VMware Fusion

A virtualization application that supports running Windows 10 as a virtual machine on macOS.

Software Provisioning



Google Compute Engine

An Infrastructure-as-a-Service virtual machine offering that allows customers to use virtual machines in the cloud.



Microsoft Azure Automation

A tool that can be used to automate the provisioning and configuration of virtual machines, databases, and other resources in Azure.



Proxmox Virtual Environment

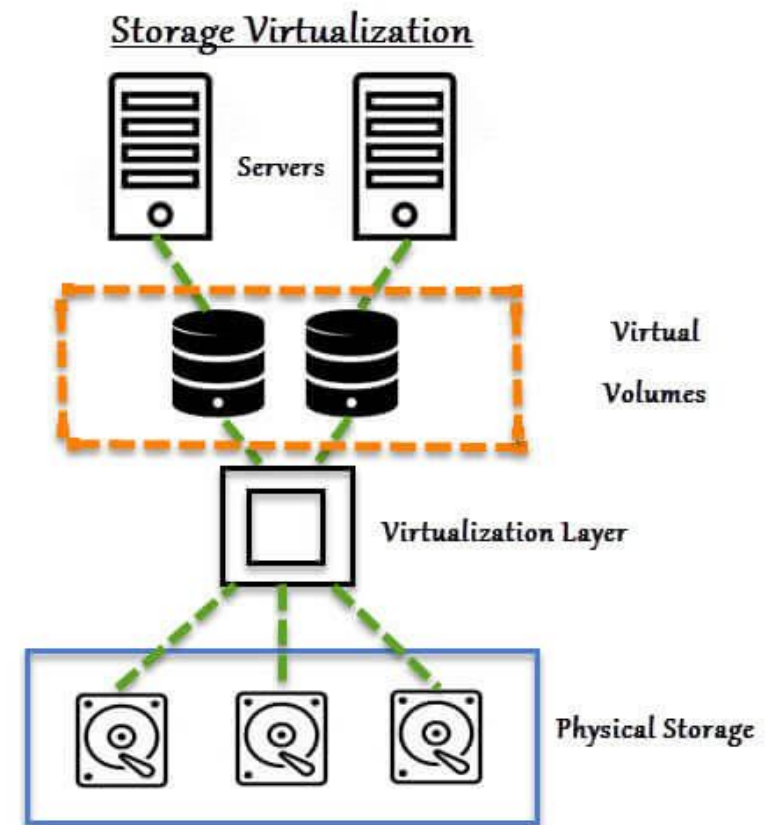
An open source virtualization solution for dedicated servers that combines KVM and OpenVZ based virtualization.

Virtualization Storage

- Storage virtualization is a technology that allows users to access physical storage resources as a single, logical pool, rather than addressing individual storage devices

Types

1. Block level storage virtualization (SAN Storage Area Network)
2. File level storage virtualization (Network Attached Storage)



Virtualization Storage

It simplifies storage management and provides a number of benefits, including:

- **Centralized management:** Storage virtualization allows for centralized management of storage resources.
- **Resource utilization:** Storage virtualization allows for the aggregation of multiple storage devices into pools, which can help with resource utilization.
- **On-demand resource management:** Storage virtualization allows for the on-demand addition and deletion of storage resources.
- **Performance and security:** Storage virtualization can improve the performance and security of storage environments.
- **Compatibility:** Storage virtualization can reduce compatibility concerns.
- **Cost and operational efficiencies:** Storage virtualization can offer significant cost and operational efficiencies over "bare metal" storage.

Hardware Provisioning

Hardware provisioning in virtualization is the process of setting up IT infrastructure, including hardware and virtual machines, and making them available to users and systems. It can help organizations scale their IT infrastructure and adapt to changing needs.

Here are some things to know about hardware provisioning in virtualization:

Hardware virtualization

- Also known as server virtualization, this process creates an abstraction layer between the software and the underlying hardware. It can improve hardware utilization and increase system efficiency.

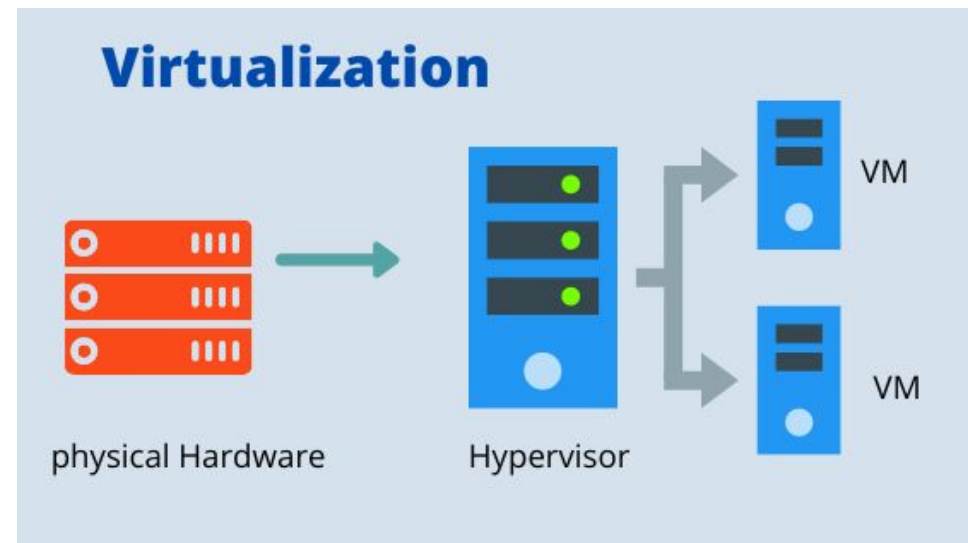
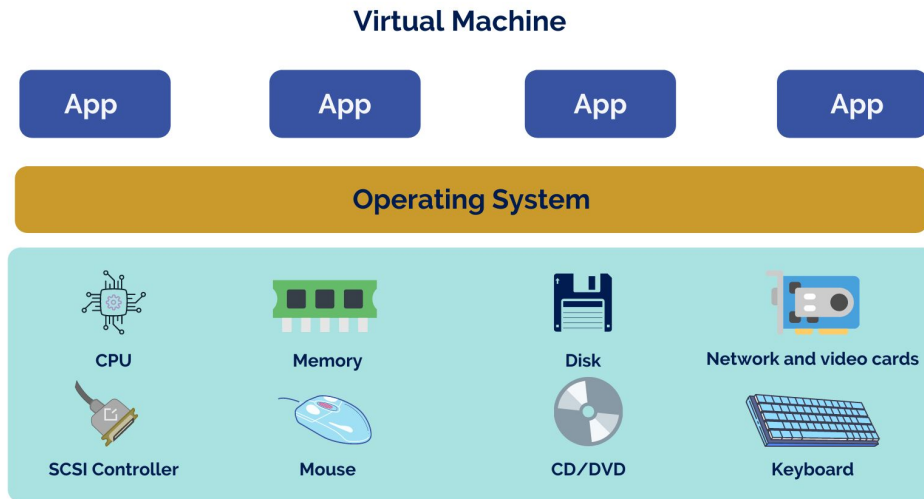
Provisioning methods

- Some methods for provisioning virtual machines include creating a template or cloning a virtual machine.

Tools for provisioning

- *Foreman* is a tool that can help with provisioning, configuring, and monitoring virtual and physical servers

Hardware Provisioning



Taking Virtualization into Cloud Considerations

- **Security:** The hypervisor can be compromised, which can negatively impact all virtual machines (VMs).
- **Support:** Who will provide support and how it will integrate with existing systems.
- **Costs:** Operational expenditures, resource management, and scalability.
- **Privacy:** How to protect personal information and identity from unauthorized users.
- **Scalability:** How to scale computing resources up or down without making major investments.
- **Flexibility:** How to pay for cloud resources based on usage.

Taking Virtualization into Cloud Considerations

The virtualized environment must be as follows:

- The virtualized services offered must be secure.
- The virtualized services must be backed up and recovered as though they're physical systems.
- These resources need to have workload management, workflow, provisioning, and load balancing at the foundation to support the required type of customer experience.
- The underpinning for the majority of high-performing clouds is a virtualized infrastructure. Virtualization has been in data centres for several years as a successful IT strategy for consolidating servers.
- Used more broadly to pool infrastructure resources, virtualization can also provide the basic building blocks for the cloud environment to enhance agility and flexibility.

VMWare Applications

- VMware App Volumes is a real-time application delivery system that enterprises can use to dynamically deliver and manage applications.
- Traditional Windows system – applications, settings, user's profile, user data are tightly coupled with the OS, and the overall user experience—productivity, preferences, and licensing—is associated with a single system.
- App Volumes application model separates out IT managed applications, user profiles, and user-installed applications into portable containers.

VMWare Applications



VMWare Applications

App Volumes Components

	APP VOLUMES MANAGER	• Dashboard console for assignments and configuration • App Volumes agent brokering • Assignment of applications and writable volumes
	APP VOLUMES AGENT	• Redirects file system and registry write operations to writable volume (if required) • Checks in with App Volumes Manager
	PROVISIONING VM	• Clean Windows virtual machine with App Volumes Agent • Snapshots used for rollback • Provision / update applications to Package
	Package	• Read-only volume containing applications • One Package : multiple users • Policy-driven virtualization
	WRITABLE VOLUME	• One writable volume : one user • Policy-driven application virtualization

Cloud Terms

- **Cloud Client**

It includes Web browsers, Mobile Apps, Thin Clients, Terminal etc.

- **Cloud Computing Provider**



- **Cloud Platform**



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THANK YOU